Microwave-assisted digestion, fluorescence staining and image analysis for microplastic quantification: a focused scoping review anchored on two group methods

Review type: focused scoping review with a documented, reproducible search (single reviewer) **Search date:** 2026-10-20 **Databases searched:** OpenAlex, Crossref, Europe PMC, PubMed, arXiv (preprint index) **Anchor papers:** 2 (the group's own prior art, retrieved by DOI) **Studies included and extracted:** 287 (abstract available for 257; a quantitative result was extractable for 147) **Context set reported but not extracted:** 235 **Protocol registration:** none (not registered; a deviation from systematic-review practice, disclosed in §8) **Working data:** `runs/litreview_v2/` — search manifest, raw responses, citation-chain manifest, screening table, PRISMA counts, extraction table, citation report **Relationship to v1:** `microplastics_review_v1.md` covered the whole field (identification / shape / polymer type / quantification). This review is **narrower and deeper**: it is scoped to the method stack the group actually works with, and it retrieves the two papers that describe that stack, which v1 did not find.

Abstract

This review synthesises 287 studies retrieved for a specific methodological stack — microwave-assisted or chemical digestion of the matrix, Nile Red fluorescence staining, and image analysis with machine learning — in sediment, soil, fish and other matrices the group works in, anchored on the group's own two published methods. The anchors are the point of the exercise: v1's broad field sweep did not retrieve either paper, because neither title nor abstract uses the generic field terms v1's queries used. Retrieving them by DOI and then expanding them by citation chaining in both directions changes what the review is about, and it exposes the group's own work as a node in a citation network rather than as an isolated result. The findings that matter for the pipeline are these. (i) **The polymer-identification claim is the contested one, and the closest prior art is not what the group may assume.** A Nile Red machine-learning study (2022) reports 95.8 % accuracy for plastic-vs-natural origin and 88.1 % for polymer type, with 92.7 % detection and 80 % identification on spiked environmental samples — but on particles of 50–1200 µm, using RGB rather than a calibrated image stack, and with a known solvatochromic basis that makes PE/PP separation the structurally hard case. (ii) **Detection probability, not model accuracy, is the binding physical limit**: the 2026 automated fluorescence system reports 50 % detection at 40 µm and 90 % at 100 µm, which means a count below ~100 µm is a censored observation, not a measurement. (iii) **Digestion is not a neutral step**: oxidation protocols can alter polymer structure by hydrolysis and can fuse particles at high temperature, so the group's 60 °C ceiling is a substantive finding rather than a convenience. (iv) **Shape taxonomy is now settled and should be adopted, not invented**: a 2025 deep-learning study applies the nine validated categories (fiber, rod, ellipse, oval, sphere, quadrilateral, triangle, free-form, unidentifiable) as reference shapes, giving the group a prior to score against instead of a two-way fibre/fragment split. (v) **Uncertainty reporting remains the field's weak flank**: 22 of 287 extracted studies report uncertainty and 4 are interlaboratory, so a pipeline that reports its own uncertainty would be ahead of the literature rather than merely compliant with it.

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1. Introduction

1.1 Background

Two published methods define what the group does. **Perez et al. (2024)**, in *PLOS ONE*, determine PET microplastics in sediment: HNO_3/HCl 3:1, microwave-assisted at 60 °C, 0.45 µm membrane, Nile Red at 10 mg/L, with 97 % recovery and a stated counting limit of particles larger than 0.066 mm. **Ferrada et al. (2024)**, in the *Journal of the Chilean Chemical Society*, determine polyethylene in *Merluccius gayi* fish: HNO_3/H_2O_2 9:1, microwave-assisted at 60 °C for 1 h, Nile Red at 2 %, with recoveries over 98 % and a demonstration that AI-based selective identification is possible. Between them they cover digestion, staining, imaging and classification — the full stack.

The 2026 FONDECYT proposal built on these two papers but departed from them in the parameters that determine the signal the classifier reads, and overstated what a preliminary dataset of 9,300 particles supports. A new proposal needs a review that is specific to the stack, and that places the group's own work in its context.

1.2 Why a focused review, and why anchored

A field-wide review (v1) is the wrong instrument for a proposal that will be judged on a specific method. It spends its synthesis on identification by FTIR, on nanoplastic toxicology and on regulatory context that the proposal will not use. Worse, it did not retrieve the group's own papers: neither title contains the generic terms the field-wide queries used, and both fell outside the screening criteria that admitted 60 studies from 4,175 records. That is a failure of the search, not of the field.

v2 therefore does two things differently. It **retrieves the anchors explicitly by DOI** and admits them by design. And it **expands each anchor by citation chaining in both directions** — the works that cite it, and the works it cites — because a paper that cites or is cited by a group method paper shares its method stack, which keyword search on broad field terms dilutes. Chaining found 33 records, of which 33 survive into the extracted set.

1.3 Objectives and review question

Question. For the method stack of microwave-assisted or chemical digestion → Nile Red fluorescence staining → image analysis with machine learning, applied to sediment, soil and fish matrices: what does the literature establish about (a) matrix digestion and polymer survival, (b) staining specificity and its confounds, (c) automated detection and classification performance, (d) shape and size reporting, and (e) quantification with its uncertainty — and what does that imply for a group that has already published two methods in this stack and intends to propose a third?

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2. Methods

2.1 Eligibility criteria

Population. Studies that determine, detect, count, size or classify microplastics in environmental or food matrices, or that develop, validate or harmonise methods for doing so.

Intervention / exposure. A sample-treatment method (digestion, density separation, filtration) or a detection/identification method (fluorescence staining, spectroscopy, image analysis with machine learning), or a study of one of these.

Comparator / reference. Any other method family, or the same method under different conditions (polymer, matrix, digestion chemistry, instrument).

Outcomes. Recovery, selectivity, detection limit, false-positive rate, classification accuracy, shape distribution, size distribution, uncertainty, interlaboratory agreement.

Study design. Methodological studies: development, validation, comparison, harmonisation, standardisation, reference-methodology work, or a review of methods. Occurrence studies that report concentrations without contributing methodology are out of scope, as are toxicological, risk-assessment and policy studies.

Window. 2015–2026 inclusive, with a documented exception: the two anchor papers and the records reached by citation chaining from them are admitted regardless of year, because chaining is itself the evidence of direct relatedness.

2.2 Information sources

OpenAlex, Crossref, Europe PMC (including its preprint index), PubMed and arXiv. All five are open and require no key. **parallel-web was unavailable** (no `PARALLEL_API_KEY` in this environment), so the scripted multi-database search is the whole of the discovery step; that is a limitation and is disclosed in §8.

2.3 Search strategy

Three stages, all scripted and all recorded in `sources/manifest.json` and `sources/manifest_chain.json` with the query, the database, the retrieval date, the reported hit count and the status of every call.

Stage 0 — anchors by DOI. The two group papers retrieved directly by DOI, with their OpenAlex work IDs recorded.

Stage 1 — citation chaining, both directions. For each anchor: the works citing it (OpenAlex `cites:<work_id>`) and the works it cites (OpenAlex `referenced_works`, with Crossref's reference list as a fallback where OpenAlex is thinly indexed — which it is for the Chilean journal, whose record carries an empty reference list). The forward filter requires the OpenAlex

work ID rather than the DOI; the first attempt used the DOI and OpenAlex returned HTTP 400, which is recorded in the manifest as a failed call rather than silently dropped.

Stage 2 — themed queries. 45 database-query combinations across five themes scoped to the stack: digestion, staining, image analysis, quantification/QA, and matrices. Queries are listed in Appendix A.

Carryover. v1's *curated* set of 60 already-screened studies is carried in as a documented source. v1's raw 4,175-record pool is deliberately **not** carried: re-screening an already-screened pool re-admits the occurrence studies v1 already excluded, and the two reviews are meant to be cumulative, not to re-litigate the same corpus.

2.4 Selection process

Deduplication by DOI, then by normalised title for records without a DOI. Screening is deterministic and rule-based, with a recorded reason for every exclusion:

1. **Topic** — the title must contain a microplastics term. (Title-scoped on purpose: an earlier version of the screen matched the method stack anywhere in title *and* abstract and admitted 854 records, most of them occurrence studies that mention "sediment" somewhere in the abstract while reporting concentrations. Requiring the signal in the *title* selects studies whose own subject is method work.)
2. **Method focus** — the title must contain a method-development signal; the abstract must confirm one.
3. **Stack and matrix** — the study must touch either a matrix the group works in (sediment, soil, fish/biota, water, and the reference-material and inter-laboratory literature) or a direct method pillar (digestion, staining, image analysis).
4. **Window** — 2015–2026, with the anchor/chain exception.
5. **Tiering** — tier 1 (extracted): at least one direct method pillar (digestion, staining, image analysis) **and** either a group matrix or a standards-setting role (reference methodology, reference material, harmonisation or standardisation, inter-laboratory study, calibration or reproducibility protocol). Tier 2 (context): meets the scope but outside that intersection; reported in the flow, citable, not extracted.

The standards-setting arm of that rule matters. A reference methodology does not name a matrix, so a matrix-only rule excluded exactly the documents the group's own method must be written against — the Nile Red protocol-harmonisation study and the reference methodology for particle size distribution. Adding the arm moved 42 such studies into the extracted set.

No full text was retrieved or assessed. The consequence is stated in §8 and in the extraction table's `extraction_basis` column, which marks every row as `rule-derived from title+abstract` except the two hand-extracted anchors.

2.5 Data extraction

For each included study: citation, DOI, anchor status, retrieval route, stack pillars, evidence type, scope/matrix, method family, the first quantitative result recoverable from the abstract, and four quality flags. The fields are **rule-derived from the title and abstract** by named, auditable regexes, so a reader who disagrees with a cell can see which rule produced it. The two anchors were hand-extracted.

2.6 Quality appraisal

No formal risk-of-bias instrument (RoB 2, AMSTAR 2) applies to a methodological scoping review of this kind, so four domain-appropriate flags were recorded per study: reports validation/accuracy/recovery; reports uncertainty or confidence; interlaboratory design; sample size reported. Results in §3.3.

2.7 Synthesis

Thematic — by the four stack pillars, not study by study. The two anchors are treated as a focal point rather than as two more rows, because the review's purpose is to locate the group's own work relative to its method stack.

2.8 Deviations from protocol

1. **Not registered.** No PROSPERO or OSF registration. Disclosed.
2. **Single reviewer.** No independent duplicate screening or full-text eligibility check. Disclosed, and the reason the extraction table marks every non-anchor row as rule-derived.
3. **No full-text assessment.** Eligibility was judged from title, abstract and metadata.

4. **parallel-web unavailable**, so the scripted search is the entire discovery step.
5. **5 search calls failed**: 4 arXiv calls (HTTP 429 or read timeout, i.e. rate limiting) and the forward chain's first attempt (HTTP 400, because the OpenAlex `cites` filter needs a work ID rather than a DOI). All are recorded in the manifest rather than dropped silently. The arXiv failures affect preprint coverage for T1 and T2 queries; the corrected chaining stage recovered the forward chain for both anchors.

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3. Results

3.1 Study selection (PRISMA)

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Identification
  records from the v2 search (themed + anchors)        4,636
  records carried from v1's curated set                   60
  -------------------------------------------------------
  records total                                        4,696
  duplicates removed                                   1,781
  -------------------------------------------------------
  records screened                                     2,915

Screening
  records excluded                                     2,393
    off-topic: no microplastics term in title+abstract         1,428
    off-topic: no microplastics term in title                     96
    not method-focused (title-scoped signal absent)               510
    not method-focused (no method signal in abstract)              42
    no group matrix and no direct method pillar                   136
    no method-stack pillar present                                120
    outside 2015-2026 window                                       61
  -------------------------------------------------------
  records after screening                              522

Eligibility
  tier 1 — core, extracted                             287
  tier 2 — context, reported not extracted             235
```

Retrieval route of the extracted set: 2 by DOI, 33 by citation chain, 252 by themed search. Both anchors are in tier 1.

3.2 Characteristics of included studies

Axis	Distribution
Method family	fluorescence_staining 108, chemical_digestion 90, image_analysis_ml 63, other 17, microwave_digestion 4, filtration 2, density_separation 1, pyrolysis_gcms 1, spectroscopy_other 1
Evidence type	protocol_harmonisation 106, method_comparison 43, method_validation 41, other 40, method_development 38, review 14, application 5
Scope / matrix	sediment 94, water 76, fish_biota 70, not stated 26, reference_material 8, food 7, plastic_product 3, air_dust 3
Year	the corpus is recent-weighted: 179 of 287 studies are from 2024–2026, and 152 are from 2025–2026

Method family is dominated by fluorescence staining (108 studies) and chemical digestion (90), with image analysis/ML at 63 and microwave-specific work at 4. That last number is the substantive finding of the whole review: **the microwave-assisted extraction literature is thin.** The group's two papers are the most-cited microwave-digestion method

papers in this set.

3.3 Quality appraisal

Flag	Studies (of 287)
Reports validation / accuracy / recovery	183
Reports uncertainty or confidence	22
Interlaboratory design	4
Sample size reported	30

Interpretation. Validation is common; **uncertainty reporting is not** (22 of 287). Interlaboratory work is rare (4 studies) and, as in v1, carries most of the field's evidence about reliability. The asymmetry — abundant single-laboratory validation, scarce uncertainty and cross-laboratory evidence — is itself a finding, and it is an opening: a pipeline that publishes its own uncertainty would be ahead of this literature rather than merely compliant with it.

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4. Synthesis by theme

4.1 Matrix digestion: a method step that changes the analyte

The stack begins with removing the matrix, and the corpus is consistent about how consequential that is. Oxidation protocols "alter the nature of polymers through hydrolysis and often do not lead to a complete matrix removal" (Pfohl et al. 2021), which is the single most important caveat for a proposal that plans to measure polymer recovery. Fenton pretreatment for organic-rich sediment "was found to be efficient in reducing organic matter content, compatible with later FTIR for polymer identification, although it did affect the size of polyethylene and PET" (Vermeiren et al. 2020) — that is, the matrix-removal step changes the *size* of the particles being measured. A systematic evaluation of digestion protocols exists precisely because the choice of digestion chemistry determines both the recovery and the size distribution reported.

This is where the group's own 60 °C ceiling becomes a **substantive methodological claim rather than a convenience**: the PLOS ONE paper reports that above 120 °C PET recovery fell because particles fused, and at 60 °C recovery was 97 %. The 2024 J. Chil. Chem. Soc. paper extends the same 60 °C logic to an acid/oxidant route in fish tissue with recoveries over 98 %, and additionally shows that lyophilization — a step common in the literature — is unnecessary and itself causes losses (≈65 % recovery).

Implication for the pipeline. Recovery and polymer survival are not a validation afterthought; they are parameters that must be fixed, reported and defended. A proposal that says "microwave digestion at ≤80 °C" (the 2026 draft's wording) is claiming a protocol *between* two published ones. Either restate one published protocol precisely, or make the optimisation of the conditions an explicit objective with pre-declared acceptance criteria. The 2026 draft did neither, and in doing so cited each published paper for a method the other describes.

4.2 Staining: the method is a screen, and its confounds are known

Nile Red's mechanism is stated plainly in the corpus: "the solvatochromic nature of Nile Red also offers the possibility of plastic categorisation based on surface polarity characteristics of identified particles" (Maes et al. 2017, *Scientific Reports*). That single sentence is the whole polymer-identification question in one line — the dye reports **polarity**, so polymer assignment is an inference from colour, and two polymers of similar polarity are structurally confusable. It also explains why the group's own 2024 paper reports PE as bright yellow while its 2024 PLOS ONE paper could not make colour do the work.

The corpus also carries the two things a proposal must handle. **False positives are pervasive**: 47 of the 287 extracted studies mention interference, false positives or organic-matter effects, and the standard remedy in the literature is a second stain or an enzymatic step to remove biological material. And **staining conditions are not standardised** — a 2025 study in the corpus is explicitly about harmonising Nile Red staining protocols (Ho and Masura 2025), which is a direct indication that the field recognises its protocols are not comparable across laboratories. The remedy is not merely aspirational: the corpus also carries a **reference methodology for microplastic particle size distribution analysis** (Richter et al. 2022), which is exactly the kind of document the group's own method should be written against.

Implication for the pipeline. A count from Nile Red-stained, fluorescence-imaged particles is a count of *stained particles* until a confounder control says otherwise. Blanks, the substrate that produces false positives, and a second stain are not optional extras; they are the difference between a count and a measurement. The 2026 draft proposed "rinsed with distilled water" as its contamination control, which is not a protocol.

4.3 Automated detection and classification: the closest prior art, and what it actually supports

This is the theme where anchoring changes the answer, because the closest prior art to the group's own approach is a paper a field-wide keyword search dilutes but a citation chain surfaces.

The closest prior art. Meyers et al. (2022), in *Science of the Total Environment*, combine Nile Red staining with RGB data from photographs to train two supervised models: a **Plastic Detection Model** and a **Polymer Identification Model**. It reports 95.8 % accuracy for plastic-versus-natural origin and 88.1 % for polymer type, and 92.7 % detection and 80 % identification on spiked environmental samples after laboratory processing. The pipeline is: stain, photograph, extract RGB, classify. That is *architecturally the same idea* as the group's descriptor-based polymer identification model.

But read the conditions. The particles are 50–1200 µm. The input is an RGB photograph, not a calibrated image stack with a recorded µm/px scale. The particles are spiked standards processed in the laboratory, not environmental samples with a confounder load. And the polymer question is exactly the one the dye's solvatochromism makes hard.

The physical limit. Tanoiri et al. (2026) are the reference point in this corpus for what an automated fluorescence system can actually do: **detection probability 50 % at 40 µm and 90 % at 100 µm**. Those numbers are the most consequential in this review, because they are *not* model properties. They are properties of the optics and the dye, and they bound what any classifier can achieve. A model reporting 95 % accuracy on particles it can only see 50 % of the time is reporting a conditional accuracy over a detection-limited subset, and the distinction has to be made explicit in any proposal or paper.

The group's own position. The measured per-class recall in the group's pipeline is PET 0.98, PP 0.70, LDPE 0.54. The 2022 study's 88.1 % polymer accuracy is *higher* than the group's PP and LDPE figures. That is not a reproach to the group's pipeline — it is a different setting, a different size range and a different label space — but it does mean the proposal cannot claim novelty on "automated polymer identification by Nile Red", because it was demonstrated in 2022. The defensible novelty is elsewhere, and §6 says where.

4.4 Shape and size: a taxonomy exists, and it should be adopted

The corpus contains a 2025 deep-learning study (Liu et al. 2025) that applies the nine validated shape categories — fiber, rod, ellipse, oval, sphere, quadrilateral, triangle, free-form, unidentifiable — as reference shapes for hyperspectral-imaged microplastics. Those are Liu et al.'s categories, and their presence here means the group can score against a published, expert-validated taxonomy rather than inventing a two-way fibre/fragment split.

The cost of adopting them should be stated plainly: the group's current `morfotipo` is a two-way split with a 1.1 % fibre rate over 6,638 measured particles. Adopting the nine categories **will change the measured shape distribution**, so it is a results-affecting decision, not a cosmetic one.

For size, the corpus is consistent that maximum Feret's diameter is the reporting quantity, and the 2026 "NRmachine" study's size-dependent detection curve gives the complementary half: what you *see* depends on the size, so a size distribution and a detection probability have to be published together or the count is uninterpretable.

4.5 Quantification and uncertainty: the field's weak flank, and the group's opportunity

Recovery is reported often (183 of 287 studies report validation, accuracy or recovery) and is usually good — the corpus contains 93 ± 10 % recovery from mussel tissue by NaOH and enzyme digestion (Catarino et al. 2016), 75 % and 77 % recovery for pretreatment and infrared imaging in a validated imaging-FTIR method (Uurasjärvi et al. 2021), and > 98 % for the group's own microwave-assisted oxidant route. **Uncertainty is reported rarely** (22 of 287). Detection limits appear in only a handful of studies, and only 4 are interlaboratory.

This asymmetry is the opportunity, and it is a concrete one. The v1 review established that no fully validated quantification method exists in the field (Vitali et al. 2023) and that interlaboratory agreement on polymer type ranges from 53 % to 100 % (van Mourik et al. 2021). v2 shows the same pattern inside the group's own niche, and shows that it is *not* because the field is full of careless work — it is full of careful single-laboratory recovery studies and almost no uncertainty.

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5. Standards and regulatory context

The relevant standards instruments are the ISO 24187:2023 series (microplastics in the environment) and the VAMAS prestandardisation platform, alongside national guidance documents in the corpus (an Arctic monitoring guidelines document appears among the extracted set). The corpus's own harmonisation literature — the 2025 Nile Red protocol harmonisation study, the inter-laboratory extraction comparisons, the reference-material work — is where a proposal should look for the *shape* of a validation package, because the group is proposing to build one.

For the specific claim the group's 2024 papers support — a microwave-assisted, Nile Red, consumer-camera method — the relevant precedent in the corpus is Richter et al. (2022)'s **reference methodology for microplastic particle size distribution analysis**, and the harmonisation effort of Ho and Masura (2025) for Nile Red staining protocols.

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6. What this means for the next proposal

The novelty claim must move. "Automated polymer identification by Nile Red" was demonstrated in 2022 at 88.1 % accuracy. The defensible contributions are the ones the corpus shows nobody has done: a **detection-probability-vs-size model** for the group's own optical stack, so that counts are censored observations with a stated detection limit; **recovery and polymer survival reported per polymer per matrix per digestion chemistry** rather than as a single headline percentage; **the unassignable fraction reported as a result**, which follows directly from the dye's solvatochromism; and **the synthetic-to-real-image performance drop** for a learned detector, which no study in this corpus reports.

The preliminary data must be real. The corpus contains field-plate-independent method-development work, but no study trains only on synthetic images and reports the drop on real filters. The group's M007 pipeline is currently a synthetic-corpus comparison whose committed verdict is an honest REVIEW with 11 of 15 gates failing. That is a legitimate *methods* contribution — pre-declared gates, honest negative results — and it is the kind of thing this field does not have. But it is not a performance claim, and a proposal should present it as the former.

The polymer scope must be decided. The proposal's target of PE, PET, PP, PVC, PA is not the code's PET, PP, LDPE, and not either paper's scope (PET in sediment; PE in fish). The literature says the dye gives a polarity class, that polymer identity is the confusable case, and that identity belongs to µFTIR or Raman. The honest and achievable claim is *counting and sizing with a polarity-based classification and explicit abstention, with a spectroscopy-confirmed subset*.

The method must be stated as one of the two published protocols, not as a blend. HNO_3/HCl 3:1 at 60 °C with Nile Red 10 mg/L, or HNO_3/H_2O_2 9:1 at 60 °C for 1 h with Nile Red 2 % — each with its published recovery, its published matrix and its published caveats. A third protocol should be a stated objective with pre-declared acceptance criteria, not an undeclared midpoint.

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7. Evidence gaps

1. **Detection probability is not modelled.** The corpus contains one study that reports it (50 % at 40 µm, 90 % at 100 µm). No study in the extracted set propagates it into a count's uncertainty.
2. **No synthetic-to-real-image drop.** No extracted study trains only on synthetic images and reports the performance drop on real filters.
3. **The microwave-assisted extraction literature is thin.** 4 of 287 extracted studies are microwave-specific, and the two most-cited of them are the group's own.
4. **Uncertainty is reported in 22 of 287 studies**, and interlaboratory design in 4.
5. **Staining protocols are not harmonised** for Nile Red in the matrices the group works in, and a harmonisation study appeared only in 2025.
6. **Size-dependent and matrix-dependent recovery are not reported together**, so a recovery figure cannot be transferred between matrices.

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8. Limitations

1. **No full-text assessment.** Eligibility was judged from title, abstract and metadata; 0 full-text reports were retrieved. The extraction table's `extraction_basis` column marks every non-anchor row `rule-derived from title+abstract`. Any

statement about a specific study's *method detail* beyond what its abstract states is therefore not supported by this review.
2. **Single reviewer.** No duplicate screening, no independent full-text eligibility check, and no inter-reviewer disagreement to report. v1 had the same limitation; v2 adds that the screening rules were written, applied and revised by the same person, so the eligibility boundary is less independent than a two-reviewer screen would be.
3. **parallel-web unavailable**, so scripted search is the entire discovery step and no supplementary grey-literature discovery was performed.
4. **Five arXiv calls failed** with HTTP 429 / timeouts. Preprint coverage for the T1 and T2 queries is therefore incomplete.
5. **The forward citation chain was initially broken** (HTTP 400: the filter needs an OpenAlex work ID, not a DOI) and was recovered in a separate recorded stage. The final chain totals are 9 forward + 23 backward for A1 and 3 forward + 0 backward for A2; the second anchor has a sparse citation network, partly because the journal is thinly indexed in OpenAlex.
6. **Tier 2 (235 studies) was not extracted**, by design. It is reported in the flow and citable, but nothing in §4 rests on it.
7. **Extraction of the two anchors was by hand from the published PDFs** held in the project's `published-papers/` folder; the corpus's abstracts were not the source for those two rows.
8. **Citation verification checks DOI registration and metadata, not whether a source supports a claim.** Every claim attributed to a specific study in §4 was taken from that study's own abstract text as retrieved, and any claim to be used in a Methods section must be re-checked against the full text.

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9. Conclusion

Scoping a review to a method stack, and anchoring it on the group's own two papers, changes the conclusions in three ways that a field-wide sweep missed. First, **the polymer-identification novelty the 2026 draft assumed does not exist** — a 2022 study built the same stain-and-classify architecture and reported higher per-class accuracy than the group's own pipeline achieves for PP and LDPE. Second, **the real ceiling is detection probability, not classifier accuracy**: the 50 %-at-40 µm figure is a property of optics and dye, and no model can exceed it, which reframes what a preliminary-data section should claim. Third, **the field's genuine weakness is uncertainty and harmonisation, not capability** — 22 of 287 studies report uncertainty and 4 are interlaboratory, in a niche where recovery is routinely 90 %+ and the group's own microwave routes exceed 97 %.

That combination points to a specific, defensible next proposal: take the two validated microwave protocols, extend them to a matrix suite with per-polymer recovery and survival, add a detection-probability model so counts become censored observations with stated limits, report the polarity-based classification with its abstention fraction as a result, and confirm a subset spectroscopically. The method is published, the recovery is high, the optics are cheap, and the quantification — the part nobody has done — is exactly where the group's software platform and pre-registered comparison methodology are the contribution.

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Appendix A — Search strings and per-database hits

Full manifest with every query, database, retrieval date, reported hit count, retrieved count and status: `runs/litreview_v2/sources/manifest.json` (themed + anchors) and `runs/litreview_v2/sources/manifest_chain.json` (citation chaining).

A.1 Themed queries

Theme	Query
T1 digestion	microwave assisted digestion microplastics extraction recovery sediment
T1 digestion	acid digestion hydrogen peroxide microplastics sample preparation matrix
T2 staining	Nile Red staining microplastics fluorescence protocol recovery specificity
T2 staining	Nile Red microplastics false positive natural organic material selectivity
T3 image analysis	machine learning image analysis microplastics detection counting classification
T3 image analysis	deep learning segmentation microplastic particles fluorescence microscopy
T4 quantification/QA	microplastics quantification detection limit recovery blank sediment fish
T4 quantification/QA	microplastics interlaboratory comparison harmonization quality assurance quantification
T5 matrices	microplastics sediment soil fish tissue digestion quantification monitoring

Each query was run against all five databases: 45 database-query combinations from 9 themed queries. Total search calls including the anchor retrievals and the citation-chaining stage: 57, of which 5 failed — 4 arXiv calls (HTTP 429 or read timeout) and the forward chain's first attempt (HTTP 400).

A.2 Anchor and chaining queries

Route	Query
Anchor A1	doi:10.1371/journal.pone.0314520
Anchor A2	doi:10.4067/s0717-97072024000106082
Forward chain A1	OpenAlex `cites:W4405493624`
Backward chain A1	OpenAlex `referenced_works` of W4405493624, plus the Crossref reference list
Forward chain A2	OpenAlex `cites:W4404352343`
Backward chain A2	OpenAlex `referenced_works` (empty for this record) and the Crossref reference list (empty)

Appendix B — PRISMA counts

Machine-readable: `runs/litreview_v2/prisma_v2.json`. The flow diagram in §3.1 is generated from that file, not written by hand.

Appendix C — Extraction table

`runs/litreview_v2/extraction_table_v2.csv` — 287 rows, one per included study, with the pillar coding, evidence type, scope/matrix, method family, the first quantitative result recoverable from the abstract, four quality flags, the retrieval route and the extraction basis. Fields are rule-derived; see §2.5.

Appendix D — Tooling contribution

- `runs/litreview_v2/search_v2.py` — themed multi-database search with per-call manifest
- `runs/litreview_v2/chain_v2.py` — citation chaining in both directions, with the OpenAlex work-ID correction and the Crossref reference-list fallback
- `runs/litreview_v2/screen_v2.py` — deduplication, deterministic tiered screening, PRISMA counts
- `runs/litreview_v2/fetch_v2.py` — per-study metadata and abstract retrieval
- `runs/litreview_v2/extract_v2.py` — rule-based extraction with the quality flags
- `runs/litreview_v2/sources/citation_report_v2.json` — DOI registration and metadata verification